

A trace formula for Hecke operators on Maass forms of prime level with primitive character

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Abstract

We state a Selberg trace formula for T_n , $(n, p) = 1$, on Maass cusp forms of prime level p with a primitive even character χ . The published formula with character (Booker–Lee–Strömbergsson, Theorem 1.7) covers only $n = \pm 1$, and Seymour-Howell’s formula for general n assumes trivial character. The new terms are the split hyperbolic classes, the continuous spectrum twisted by the Hecke eigenvalues of the Eisenstein series, and the parabolic classes for square n . Each term is derived up to its normalising constant, and every constant is pinned numerically against independently computed spectra at levels 11 and 7. After pinning, the formula predicts the traces at the remaining composite and square n to 10^{-8} , with no free parameter left. The status is [N]: a written proof of the normalisations is still needed. We then use it for a floating-point pilot of the uniqueness count at level 1951. The pilot says that the certified form near $\frac{1}{4}$ is the only newform in a window around it. Its ratio is 0.78, where any value below 1 suffices. It does not bear on Artin’s conjecture for ρ directly, and Artin’s conjecture for ρ remains open.

1 Setting

p is prime and χ is a primitive character mod p with $\chi(-1) = 1$. A Maass form satisfies $f(\gamma z) = \chi(d)f(z)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(p)$. The Hecke operator $T_n f(z) = n^{-1/2} \sum_{ad=n} \chi(a) \sum_{b \bmod d} f\left(\frac{az+b}{d}\right)$ has eigenvalues $\lambda_j(n)$ on the newforms u_j , with $\lambda_j = \frac{1}{4} + r_j^2$. All forms are new, since χ is primitive and p is prime. Let h be even and holomorphic in a strip, and let $g(u) = \frac{1}{2\pi} \int h(r) e^{-iru} dr$. Write $L(1, \psi_D)$ for Zagier’s L -value of a nonsquare discriminant $D = dl^2$: $L(1, \psi_D) = L(1, \psi_d) l^{-1} \prod_{q^e \parallel l} (1 + (q - \psi_d(q)) \frac{q^e - 1}{q - 1})$.

2 The formula

Theorem 1 ([N]). *For $(n, p) = 1$,*

$$\sum_j h(r_j) \lambda_j(n) = \mathcal{H}_n + \mathcal{C}_n + \mathcal{S}_n + \mathcal{Q}_n,$$

where the four terms are as follows.

Non-split classes. Here $D = t^2 - 4n$ runs over nonsquares, and $H_{t,n}(\chi)$ is the local factor of [1, Thm. 1.7] at p :

$$\mathcal{H}_n = \sum_{\substack{t \in \mathbb{Z} \\ D > 0}} H_{t,n} L(1, \psi_D) g\left(2 \log \frac{|t| + \sqrt{D}}{2\sqrt{n}}\right)$$

$$+ \sum_{D < 0} H_{t,n} L(1, \psi_D) \frac{\sqrt{|D|/n}}{\pi} \int_0^\infty \frac{g(u) \cosh(u/2)}{4 \sinh^2(u/2) + |D|/n} du.$$

Continuous spectrum. Let $\lambda(n, r) = \sum_{ad=n} \chi(d)(a/d)^{ir}$, the Hecke eigenvalue of $E_{1,\chi}$. Then

$$\begin{aligned} \mathcal{C}_n &= \frac{1}{2\pi} \int_{\mathbb{R}} h(r) \lambda(n, r) (-\psi(\tfrac{1}{2} + ir) - \psi(\tfrac{1}{2} - ir)) dr \\ &+ 2 \sum_{m \geq 2} \frac{\Lambda(m)}{m} \sum_{ad=n} \chi(d) (\bar{\chi}(m) g(\log \tfrac{a}{d} + 2 \log m) + \chi(m) g(\log \tfrac{a}{d} - 2 \log m)). \end{aligned}$$

Split classes. Here $L = \log(a/d)$ and $m = |a - d|$, and the sum runs over ordered pairs:

$$\begin{aligned} \mathcal{S}_n &= \sum_{\substack{ad=n \\ a \neq d}} \frac{\chi(a) + \chi(d)}{2} \left[x_p(m) g(L) + \tfrac{1}{2} J(L) \right], \\ J(L) &= -4 \int_{|L|}^\infty g'(u) \log \frac{1 + \sinh(u/2)/\sinh(|L|/2)}{2} du, \\ x_p(m) &= \frac{2}{m} \sum_{q|m} \varphi(q) \log q + \frac{\log p}{m} \sum_{q|m, p \nmid q} \varphi(q) + 2 \log \pi - 3 \log p. \end{aligned}$$

Square classes. Here $n = k^2$; otherwise $\mathcal{Q}_n = 0$. $R_1(g)$ is the $n = 1$ remainder:

$$\begin{aligned} \mathcal{Q}_{k^2} &= \chi(k) [R_1(g) + 2g(0) \log k] + \frac{\chi(k)}{k} I(g), \quad I(g) = -\frac{p+1}{6} \int_0^\infty \frac{g'(u)}{\sinh(u/2)} du, \\ R_1(g) &= 2 \left[(2\gamma + \log 4\pi + \tfrac{1}{2} \log 2 - \tfrac{3}{2} \log p) g(0) - \int_0^\infty \log(\sinh \tfrac{u}{2} \tanh \tfrac{u}{4}) g'(u) du \right] \\ &\quad - g(0) \log 2 + \frac{1}{\pi} \int_{\mathbb{R}} h(r) \operatorname{Re} \psi(\tfrac{1}{2} + ir) dr. \end{aligned}$$

At $n = 1$ the theorem reduces to [1, Thm. 1.7] in the case $s = e = 1$. One correction is needed there: the $\log 2$ coefficient of $g(0)$ is 1, which our transcription of the theorem garbles.

3 Where each term comes from

Continuous spectrum. For primitive χ the scattering matrix is anti-diagonal in the basis $E_{1,\chi}, E_{\chi,1}$, which is also the Hecke eigenbasis. The two series are exchanged by $s \mapsto 1 - s$, and their eigenvalues are $\lambda(n, \pm r)$. By the Maass–Selberg relations, the truncated trace of T_n on the continuous spectrum is $2 \log Y \sum_{ad=n} \chi(d) g(\log \tfrac{a}{d})$ plus $\frac{1}{2\pi} \int h(r) \lambda(n, r) \omega(r) dr$. Here ω is the log-derivative of one scattering coefficient, $\propto p^{-s} \sqrt{\pi} \Gamma(s - \tfrac{1}{2}) \Gamma(s)^{-1} L(2s - 1, \chi) / L(2s, \chi)$. The functional equation of $L(s, \chi)$ turns $\omega(\tfrac{1}{2} + ir)$ into $2 \log \pi - 3 \log p - \psi(\tfrac{1}{2} + ir) - \psi(\tfrac{1}{2} - ir)$ plus the prime sums. The constant term is then absorbed into x_p . The other three assignments of χ and $\bar{\chi}$ fail the numerical test at level 11 at order 10^{-1} .

Split classes. A split class of determinant n is conjugate to $\gamma = \begin{pmatrix} a & \beta \\ 0 & d \end{pmatrix}$, with β taken mod $m = |a - d|$. Its second fixed point $\xi = \beta/(d - a)$ has denominator q , where $q \mid m$. Classes fixing the cusp 0 are handled by conjugating with the Fricke involution, which exchanges $\chi(a)$ and $\chi(d)$. The orbital integral over $\mathbb{H}$, with horoballs of height Y removed at ∞ and at ξ , is computed in the coordinates $z \mapsto (a/d)z$. Let $\delta_\xi = (q^2 Y)^{-1}$, multiplied by p^{-1} when $p \nmid q$, be the diameter of the horoball at ξ . Then the integral is

$$\frac{2\sqrt{n}}{m} g(L) \log \frac{Y}{\delta_\xi} + J_k(L), \quad J_k(L) = \int_{\mathbb{R}} k(c(1+v^2)) \log(1+v^2) dv, \quad c = \sinh^2 \frac{L}{2}.$$

Abel inversion, together with $\int_c^t \log(x/c) \frac{dx}{\sqrt{(x-c)(t-x)}} = 2\pi \log \frac{1+\sqrt{t/c}}{2}$, gives J_k in terms of g , and its contribution is $\frac{1}{2}J(L)$ above. Summing $\log(1/\delta_\xi)$ over β produces the arithmetic part of $x_p(m)$.

Square classes. For $n = k^2$ the classes $\begin{pmatrix} k & \beta \\ 0 & k \end{pmatrix}$ are the $n = 1$ parabolic classes with y rescaled by k . This gives $\chi(k)$ times the $n = 1$ block, plus $2g(0) \log k$ from the rescaled truncation. The identity element kI gives $\chi(k)k^{-1}$ times the $n = 1$ identity term.

4 Numerical confirmation

Spectra were computed independently with a Hejhal solver (both parities, Hecke relations to 10^{-7}). The two levels are $p = 11$ with χ of order 5 (63 forms, $R \leq 10.5$) and $p = 7$ with χ of order 3 (26 forms, $R \leq 9$). The test functions are Gaussians $h(r) = e^{-r^2/2\sigma^2}$.

what	how it was fixed	check
coefficient $\frac{1}{2}$ of J	fit, $n = 2, 3, 5, 7$ at level 11	0.49998, 0.49995, 0.5004, 0.4998
constant $2 \log \pi - 3 \log p$	fit, same data	-4.90422 against -4.904214
coefficient 2 of $g(0) \log k$	fit at $n = 4$	2.0000000
$n = 6, 8, 10, 12, 14, 15$ (level 11)	prediction	$ \text{spec} - \text{geom} \leq 5 \times 10^{-7}$ (mostly $\sim 10^{-8}$)
$n = 9$ (level 11)	prediction	$\log k$ coefficient 2.0000000
level 7, order-3 χ : $n = 1, \dots, 6, 8, 9$	prediction, nothing refitted	$ \text{spec} - \text{geom} \leq 7 \times 10^{-8}$ ($\leq 1.1 \times 10^{-8}$ at $\sigma = 0.8$)
$n = 8$ at level 7 ($m = 7 = p$)	tests the $p \mid q$ branch of x_p	5.9×10^{-9}
production code, bump g , level 11, $n = 1$	comparison, $a = 6, 8, 10$	4.0, 1.3, 0.60×10^{-5} ; these track the Weyl tail beyond $R = 10.56$ (2.5, 0.78, 0.30×10^{-5}), but the ratio of 1.6-2.0 is not accounted for

5 Uniqueness of the form near $\frac{1}{4}$ at level 1951: a pilot

Take $p = 1951$, and let χ be the quintic character of the project, $\chi(n) = e^{4\pi i \text{ind}_3(n)/5}$. For every newform the numbers $t_p = \bar{\chi}(p)^3 \lambda(p)$ are real, since $\chi^5 = 1$, and the Hecke relations give $\lambda(2^e 3^f) = \chi(2^e 3^f)^3 U_e(t_2/2) U_f(t_3/2)$, with U_k the Chebyshev polynomials of the second

kind. Let $f(u) = \cos^2(\pi u/a)$ for $|u| \leq a/2$, $g = f * f$ and $h = \hat{f}^2$. Then $h \geq 0$ on $\mathbb{R}$, and $h(it) = \int g \cosh(tu) du \geq 0$, so every spectral weight is non-negative, exceptional eigenvalues included.

Proposition 2 ([T], with u_n defined spectrally). *Let $c = (c_{ab})_{a \leq A, b \leq B}$ be real. Put $u_n = \bar{\chi}(n)^3 \sum_j h(r_j) \lambda_j(n)$, the spectral sum itself, and $Q(c) = \sum_{a,b,a',b'} c_{ab} c_{a'b'} \sum_{e,f} u_{2e3f}$. Here e runs from $|a - a'|$ to $a + a'$ and f from $|b - b'|$ to $b + b'$, both in steps of 2. Let W be the set of (λ, t_2, t_3) with $|r| \leq r_W$ (real or imaginary), $|t_2 - 1| \leq \varepsilon$ and $|t_3 + \varphi| \leq \varepsilon$. If*

$$Q(c) < 2 h_{\min} \min_W \left(\sum c_{ab} U_a\left(\frac{t_2}{2}\right) U_b\left(\frac{t_3}{2}\right) \right)^2, \quad h_{\min} = \min_{|r| \leq r_W} h(r),$$

then at most one newform of level 1951 and character χ lies in W .

Proof. By the Clebsch–Gordan rule $U_a U_{a'} = \sum_e U_e$, we have $Q(c) = \sum_j h(r_j) P_j$ with $P_j = (\sum c_{ab} U_a(t_{2,j}/2) U_b(t_{3,j}/2))^2 \geq 0$. Two distinct newforms in W would each contribute at least $h_{\min} \min_W P$. $\square$

The proposition uses only positivity, so it holds as stated. Applying it requires the spectral sums u_n , and the pilot computes those from Theorem 1, which is [N], using class numbers that assume GRH. *The conclusion of the pilot is therefore [N], not [T].*

The pilot [N]. We take $a = 18$, $A = B = 2$, and the 25 traces with $n \mid 2^4 3^4$. Theorem 1 is evaluated in floating point, using 1.8×10^6 values $L(1, \psi_D)$ with $|D| \leq 2.1 \times 10^{10}$ (PARI, 70 minutes on one core).

- $\max_n |\operatorname{Im} u_n| = 4.5 \times 10^{-10}$, consistent with the reality of t_2, t_3 .
- The optimal $c = M^{-1}v$, where v is the vector of Chebyshev values at $(t_2, t_3) = (1, -\varphi)$, gives $Q = 126.36$, against the threshold $2h_{\min}(c \cdot v)^2 = 162.00$ for the certified window $(|\lambda - \frac{1}{4}| \leq 8.7 \times 10^{-10}, \varepsilon = 2 \times 10^{-13})$. The ratio is 0.780. Degrees (1, 2) alone give 0.793, and degrees (1, 1) give 0.934.
- The same c still gives a ratio below 1 for the much larger window $|\lambda - \frac{1}{4}| \leq 2.5 \times 10^{-3}$, $|t_2 - 1| \leq 0.05$, $|t_3 + \varphi| \leq 0.05$.
- Sanity: $h(0) v^\top M^{-1}v = 0.641 \in (\frac{1}{2}, 1]$, and M is positive definite ($\lambda_{\min} = 250$ for $A = B = 2$; 396 for degrees (1, 2)).

Existence of a form in the window is the eigenvalue certificate of the companion paper ([T|A]). Combined with the pilot, this says, numerically, that ϕ is the only newform in the window. It does not say that $\lambda_\phi = \frac{1}{4}$, and it does not say that $\lambda_\phi(p) = a_p(\rho)$ at every prime.

What is missing for [T].

1. A written proof of Theorem 1, with all normalisations. These are currently derived up to constants that are pinned numerically.
2. Unconditional values of $L(1, \psi_D)$. PARI's `quadclassunit` assumes GRH, and `qfbclassno` is not guaranteed. The 1.8×10^6 values would have to be recomputed by counting reduced forms. That costs $O(|D|^{1/2})$ each, which is feasible in C.
3. Interval arithmetic for the quadratures and the ψ -integral. The margin is 22%, so modest precision suffices.

References

- [1] A. R. Booker, M. Lee, A. Strömbergsson, *Twist-minimal trace formulas and the Selberg eigenvalue conjecture*, J. London Math. Soc. (2020), arXiv:1803.06016.
- [2] A. Seymour-Howell, *Rigorous computation of Maass cusp forms of squarefree level*, arXiv:2201.08760.